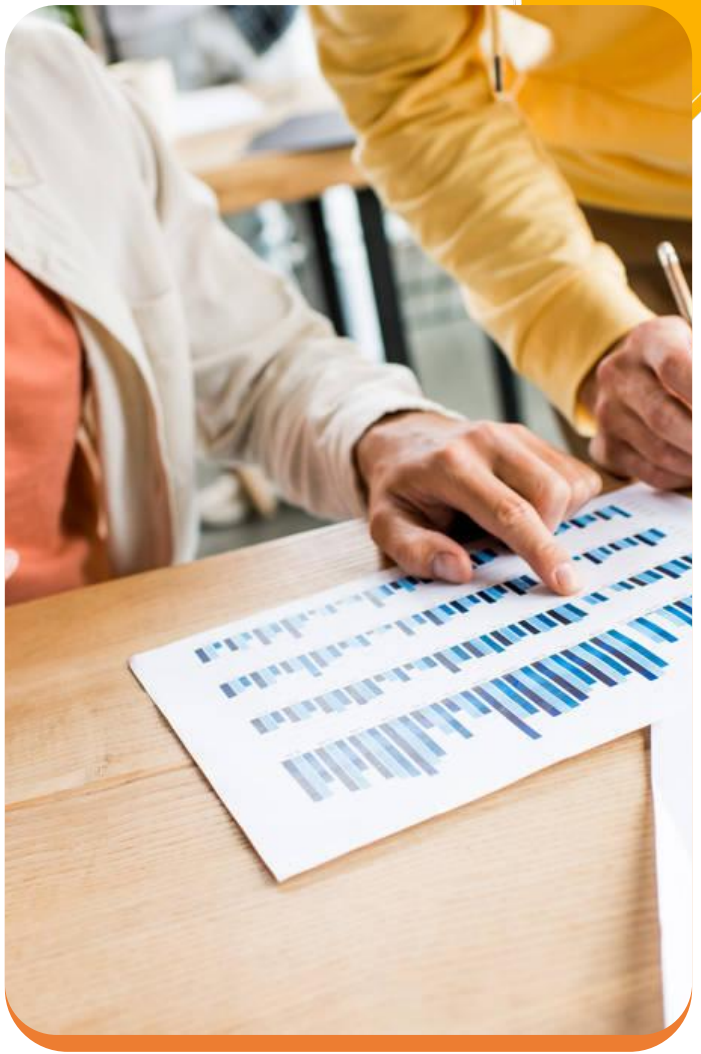
Logistic Regression Analysis



Logistic Regression Analysis

01 It estimates the probabilities of the possible outcomes using a logistic function.

02 Logistic regression is used for analyzing datasets with one or more independent variables that determine a binary outcome.

Research Questions

01 Example research question: Does a set of symptoms predict whether a patient has a certain disease?

02 Logistic regression is used to predict the presence or absence of a characteristic or outcome based on predictor variables.

Scenario Example

- Example scenario: Studying the relationship between taking a new drug and the reduction of heart attack risk.
- Independent variables: Dosage of the drug, patient age, gender, cholesterol level, and blood pressure.
- Dependent variable: Whether a heart attack occurred or not.

Outcomes in Logistic Regression Analysis

- For each observation, logistic regression models provide the probability of the observation being classified as a 'success' (e.g., having the disease, passing the exam).
- The odds of an event happening (e.g., success) vs. it not happening (e.g., failure). An odds ratio greater than 1 indicates that the event is more likely to happen as the independent variable increases.
- Based on a probability threshold (commonly 0.5), observations can be classified into one of the two categories.
- These indicate the direction and strength of the relationship between each independent variable and the log odds of the dependent variable.

$$\boxed{\log\left(\frac{p}{1-p}\right)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where:

- p is the probability of the dependent variable equaling a 'success' or '1'.
- $\frac{p}{1-p}$ is the odds of the dependent variable equaling a 'success' or '1'. It represents the ratio of the probability of success to the probability of failure.
- $\log\left(\frac{p}{1-p}\right)$ is the natural logarithm (log) of the odds of the dependent variable being a 'success' or '1'. This transformation is what makes the response variable (left-hand side) linear with respect to the independent variables.
- β_0 is the intercept term. It represents the log odds of the dependent variable equaling 1 when all the independent variables are 0.
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients of the independent variables ($X_1, X_2, \ldots, X_n$). These coefficients represent the change in the log odds of the dependent variable being 1 for a one-unit change in the corresponding independent variable, assuming all other variables are held constant.

Interpretation of Coefficients

- **Intercept (β_0)**: The log odds of the dependent variable being 1 when all independent variables are 0.
- **Slope Coefficients ($\beta_1, \beta_2, \ldots, \beta_n$)**: Each coefficient represents the change in the log odds of the dependent variable being 1 for a one-unit change in the corresponding independent variable, holding all other variables constant. A positive coefficient indicates that as the independent variable increases, the probability of the dependent variable being 1 increases. Conversely, a negative coefficient indicates that as the independent variable increases, the probability of the dependent variable being 1 decreases.

Consider a model predicting whether a patient has a certain disease (1 = yes, 0 = no) based on their age (X_1), BMI (X_2), and smoking status (X_3, where 1 = smoker and 0 = non-smoker). The logistic regression equation might look like this:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{BMI}) + \beta_3(\text{Smoking Status})$$

0.75 > 0.5 ~ 1

Interpreting the coefficients in this context:

- β_1 would indicate how the log odds of having the disease changes with a one-year increase in age, holding BMI and smoking status constant.
- β_2 would show the change in the log odds of having the disease with a one-unit increase in BMI, holding age and smoking status constant.
- β_3 would reflect how the log odds of having the disease differ between smokers and non-smokers, holding age and BMI constant.

Thanks for watching. 😄